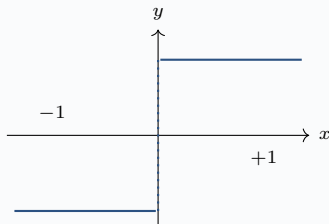
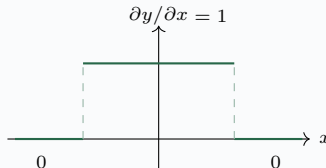


Aux A. Straight-Through Estimator ($\widetilde{\text{sign}}$)

Forward: $y = \text{sign}(x)$



Backward: $\partial y / \partial x = \mathbf{1}[|x| \leq 1]$



Forward is hard binarisation; backward is identity on the active strip $[-1, 1]$ and zero outside. The continuous shadow $\mathbf{V}$ is updated by backpropagation; the recurrence always sees $\tilde{\mathbf{V}} \in \{-1, +1\}^{V \times d}$.

Aux B. Deployment equivalent — XOR + popcount

Encode $-1 \mapsto 0$, $+1 \mapsto 1$. Then for $a, b \in \{-1, +1\}^d$:

$$\langle a, b \rangle = d - 2 \cdot \text{popcount}(\bar{a} \oplus \bar{b}).$$

$\bar{a}$:	1	0	1	1	0	1	0	1
$\bar{b}$:	1	1	1	0	0	1	1	0
$\bar{a} \oplus \bar{b}$:	0	1	0	1	0	0	1	1

popcount = 4
 $d - 2 \cdot 4 = 0$
 (orthogonal)

The 6502 inner loop: XOR two $d/8$ -byte rows, popcount the result, subtract $2 \cdot \text{popcount}$ from d . No multiplications, no floating point, no NaN reachable. For $d=256$: 32 bytes XORed and one popcount lookup per byte.